

Total No. of Questions : 10]

SEAT No. :

P-9341

[Total No. of Pages : 2

[6181]-576

B.E. (Mechanical)

**HEATING, VENTILATION & AIR CONDITIONING AND
REFRIGERATION ENGINEERING**

(2015 Pattern) (Semester - I) (402044C) (Elective - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Assume suitable data wherever necessary. Use of refrigeration table is permitted.
- 3) Use of non-programmable pocket calculator is allowed.
- 4) Draw neat diagrams wherever necessary.
- 5) Figures to the right indicate full marks.

Q1) a) Explain working of Ejector with neat sketch. Write advantages of ejector? [3]

b) Classify, cooling tower and explain hyperbolic cooling tower with neat sketch. [7]

OR

Q2) a) Explain Critical Point and represent it on phase diagram for CO₂. [3]

b) What do you mean by Capacity Controls of reciprocating compressors? [7]

Q3) a) Write significance of characteristic curves of compressors. [3]

b) Write short notes on Refrigerant Piping. [7]

OR

Q4) a) Explain with neat sketch, working of Evaporative Condensers. [5]

b) Explain the sizing process of suction line in refrigeration system. [5]

P.T.O.

- Q5)** a) Write short notes on Indoor Air Quality. [8]
b) Write a short note on decrement factor & time lag. [8]

OR

- Q6)** a) Explain the causes of Indoor air pollution. [8]
b) Explain the Indoor design parameter of air conditioning system. [8]

- Q7)** a) Write a short note on components of cooling load. [8]
b) Explain the purpose of Energy Conservation Building Code (ECBC). [8]

OR

- Q8)** a) Define: [10]
i) Space Heat Gain,
ii) Space cooling load
iii) Space heat extraction rate
iv) Coil load 5. Sol air temperature.

- b) What do you understand by space load characteristics? [6]

- Q9)** a) Explain with neat sketch Solid Desiccant Air-Conditioning. [8]
b) What are thermal storage cooling air conditioning systems? [10]

OR

- Q10)** a) What are Clean-Room Air Conditioning Systems? [8]
b) Explain with neat sketch evaporative cooling air conditioning systems. [10]
